Thermo-hydraulic study of a NEPCM micro-channel with an oscillating NACA 0012 airfoil

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Abstract

This study assesses an oscillating NACA 0012 airfoil, combined with a second fixed airfoil, and nano-encapsulated phase change materials (NEPCM) in a heated micro-channel. The unsteady Fluent model, which uses a sliding mesh and mixture properties implemented in a user-defined function, is verified in space and time and then used for a parametric study of the oscillation amplitude, the oscillation frequency and the NEPCM fraction. The airfoils raise heat transfer by a few percent but nearly double the pressure drop, so that the performance evaluation criterion remains below 0.9. NEPCM degrade performance: only a small fraction of the capsules melts, whereas their effect on viscosity applies to the whole flow.

1 Introduction

High-power-density electronic components are increasingly cooled with micro-channels. The flow in these channels is laminar: heat diffuses slowly from the wall to the core of the fluid, which limits heat transfer.

Two approaches are examined here to improve it. The first acts on the flow: an oscillating NACA 0012 airfoil generates a wake that mixes the fluid near the heated wall. The second acts on the fluid: nano-encapsulated phase change materials (NEPCM) are added to water so as to absorb latent heat as they melt.

The objective is to determine, by numerical simulation, whether these two solutions improve the trade-off between heat transfer and pressure drop, and to identify the parameters that control this trade-off.

2 Problem statement

The domain is a 2D planar channel of length $L = 15$ mm and height $H = 1$ mm ($D_h = 2H$). Two adiabatic NACA 0012 airfoils of chord $c = 0.5$ mm are placed in the channel (Fig. 1):

- **airfoil 1** oscillates about a pivot located at the quarter chord, at (6.5; 0.5) mm, with a mean angle $\alpha_0 = -10°$;
- **airfoil 2** is fixed, centred at (10; 0.375) mm and inclined at $-15°$.

The boundary conditions are given in Table 1.

Airfoil 1 follows the law

$$\theta(t) = \alpha_0 + \theta_0 \sin(2\pi f t), \qquad f = \frac{St\, u_e}{c}, \qquad \tau = \frac{1}{f}, \tag{1}$$

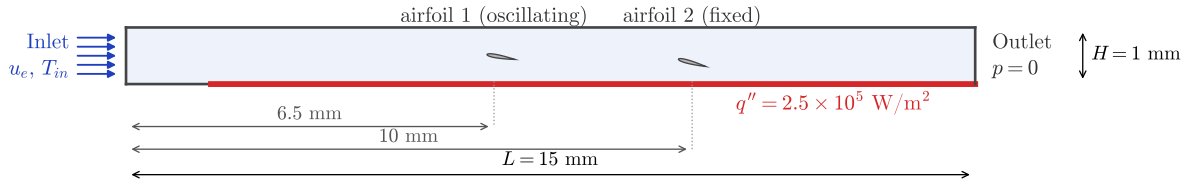


Figure 1: Computational domain and boundary conditions.

Table 1: Boundary conditions.

Boundary	Condition
Inlet	uniform velocity $u_e = 0.2232$ m/s, $T_{in} = 298.15$ K
Outlet	$p = 0$ Pa
Bottom wall, $x \geq 1.5$ mm	uniform heat flux $q'' = 2.5 \times 10^5$ W/m^2
Bottom wall, $x < 1.5$ mm	adiabatic, no slip
Top wall	adiabatic, no slip
Airfoils 1 and 2	adiabatic, no slip

where θ_0 is the amplitude, f the frequency, St the Strouhal number and τ the period.

Performance is compared, at equal velocity, with that of the smooth channel without airfoils (subscript 0):

$$PEC = \frac{Nu/Nu_0}{(\Delta p/\Delta p_0)^{1/3}}, \qquad PEC_{act} = \frac{Nu/Nu_0}{[(P_{pump} + P_{act})/P_{pump,0}]^{1/3}}, \tag{2}$$

with:

- $Nu = \dfrac{q''\, D_h}{k_w\,(\bar{T}_w - T_{in})}$: Nusselt number, where $\bar{T}_w$ is the mean temperature of the heated wall and $k_w = 0.6064$ W/m K the thermal conductivity of water at T_{in} [1];
- Δp: pressure drop between $x = 1$ and $x = 14$ mm;
- T_{max}: maximum temperature of the heated wall;
- $P_{pump} = \Delta p\, u_e\, H$: pumping power;
- $P_{act} = \dfrac{1}{\tau}\displaystyle\int_\tau M_z\,\dot{\theta}\,\mathrm{d}t$: actuation power of airfoil 1, where M_z is the moment about the pivot;
- $v_{rms} = \sqrt{\dfrac{1}{H}\displaystyle\int_0^H (v - \bar{v})^2\,\mathrm{d}y}$: transverse agitation in the section $x = 12.5$ mm, where v is the transverse velocity and $\bar{v}$ its mean over the section.

All quantities are averaged over the last three periods, once the periodic regime is established.

3 Governing equations and property models

The suspension is treated as a single homogeneous fluid; the flow is laminar, incompressible and unsteady:

$$\nabla \cdot \mathbf{u} = 0, \tag{3}$$

$$\rho\left[\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u}\right] = -\nabla p + \nabla\cdot(\mu(T)\nabla\mathbf{u}), \tag{4}$$

$$\rho\, C_p(T)\left[\frac{\partial T}{\partial t} + \mathbf{u}\cdot\nabla T\right] = \nabla\cdot(k(T)\nabla T). \tag{5}$$

Subscripts w and p denote water and capsule, and φ is the capsule volume fraction.

Density. It is constant:

$$\rho = (1-\varphi)\,\rho_w + \varphi\,\rho_p. \tag{6}$$

Thermal conductivity. The conductivity of water follows the correlation of Ramires et al. [1], and that of the mixture the Maxwell model [3]:

$$k_w(T) = 0.6064\left[-1.48445 + 4.12292\,\frac{T}{298.15} - 1.63866\left(\frac{T}{298.15}\right)^2\right], \tag{7}$$

$$k(T) = k_w(T)\,\frac{2k_w(T) + k_p + 2\varphi\big(k_p - k_w(T)\big)}{2k_w(T) + k_p - \varphi\big(k_p - k_w(T)\big)}. \tag{8}$$

Dynamic viscosity. The viscosity of water follows the formulation of Huber et al. [2], and that of the mixture the Batchelor relation [4]:

$$\mu_w(T) = 10^{-6}\left[280.68\,T_r^{-1.9} + 511.45\,T_r^{-7.7} + 61.131\,T_r^{-19.6} + 0.45903\,T_r^{-40}\right], \qquad T_r = \frac{T}{300}, \tag{9}$$

$$\mu(T) = \mu_w(T)\left(1 + 2.5\,\varphi + 6.2\,\varphi^2\right). \tag{10}$$

Heat capacity. The capsules melt between $T_1 = T_f - T_{mr}/2$ and $T_2 = T_f + T_{mr}/2$, where T_f is the melting temperature and T_{mr} the width of the melting range. Melting is modelled by a sinusoidal peak added to the heat capacity of the capsule [5]:

$$C_p(T) = \begin{cases} \omega\left[C_{p,p} + A\sin\!\left(\pi\,\dfrac{T - T_1}{T_{mr}}\right)\right] + (1-\omega)\,C_{p,w} & \text{if } T_1 \le T \le T_2, \\ \omega\,C_{p,p} + (1-\omega)\,C_{p,w} & \text{otherwise,} \end{cases} \tag{11}$$

with

$$\omega = \frac{\varphi\,\rho_p}{\rho}, \qquad A = \frac{\pi}{2}\left(\frac{h_{sf}}{T_{mr}} - C_{p,p}\right), \tag{12}$$

where ω is the capsule mass fraction, h_{sf} the energy absorbed by the capsule over the melting range and A the amplitude of the peak.

Melt fraction. The melted fraction of the capsule β and its flow-weighted average at the outlet β_s are:

$$\beta(T) = \begin{cases} 0 & T < T_1, \\ \dfrac{1}{2}\left[1 - \cos\!\left(\pi\,\dfrac{T - T_1}{T_{mr}}\right)\right] & T_1 \le T \le T_2, \\ 1 & T > T_2, \end{cases} \qquad \beta_s = \left.\frac{\displaystyle\int_0^H \rho\,u\,\beta\,\mathrm{d}y}{\displaystyle\int_0^H \rho\,u\,\mathrm{d}y}\right|_{\text{outlet}}. \tag{13}$$

These laws are implemented in a user-defined function (UDF) in Ansys Fluent 2026 R1. The equations are discretised with second-order schemes in space and time, with a time step $\Delta t = \tau/100$.

Table 2: Material properties (capsule with an n-octadecane core [6]).

Property	Water	Capsule
ρ (kg/m^3)	997.1	1033
k (W/m K)	$k_w(T)$	0.2
C_p (J/kg K)	4179	1830
μ (Pa s)	$\mu_w(T)$	–
h_{sf} (J/kg)	–	152 600
T_f; T_{mr} (K)	–	301.15; 0.5

4 Mesh

The mesh is structured; airfoil 1 lies in a rotating zone connected to the rest of the domain by a sliding interface (Fig. 2). Its quality is given in Table 3.

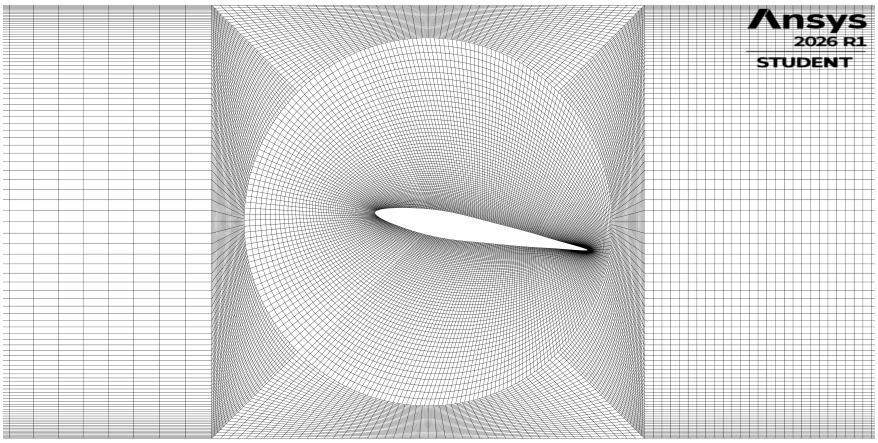


Figure 2: Mesh around airfoil 1: rotating zone and surrounding blocks.

Table 3: Quality of the medium mesh.

Metric	Value
Number of cells	85 700
Minimum orthogonal quality	0.266
Maximum skewness	0.796
Maximum aspect ratio	26.3

5 Validation of the method

The sliding-mesh method was first validated against the laminar simulations of Kurtulus for a NACA 0012 airfoil at $Re = 1000$: a static airfoil at $\alpha = 9°$ [7] and an airfoil pitching by $\pm 1°$ about $\alpha_0 = 9°$ and $20°$ [8]. The solver and schemes are those of the present study. Deviations remain below 3 %, within the digitisation uncertainty of the reference figures (Table 4).

6 Mesh and time-step convergence (GCI)

The GCI calculations [9] are performed for $\theta_0 = 10°$, $St = 0.3$ and $\varphi = 0.04$.

Table 4: Validation against the cases of Kurtulus (St: vortex-shedding Strouhal number; C_l, C_d: mean lift and drag coefficients).

Case	Quantity	Present study	Reference	Deviation
Static, $\alpha = 9°$	St	0.891	0.89 [7]	+0.2 %
Pitching, $\alpha_0 = 9°$	C_l	0.406	0.407 [8]	−0.4 %
	C_d	0.156	0.151 [8]	+3 %
Pitching, $\alpha_0 = 20°$	C_l	0.927	0.915 [8]	+1.3 %
	C_d	0.448	0.461 [8]	−2.8 %

Table 5: Spatial discretisation error ($r = 1.5$; subscripts: 1 = fine, 2 = medium, 3 = coarse).

Quantity	$\psi = Nu$	$\psi = T_{max}$ (K)	$\psi = \beta_s$
N_1; N_2; N_3		192 990; 85 700; 36 731	
ψ_1; ψ_2; ψ_3	18.673; 18.689; 18.708	359.218; 359.201; 359.170	0.1902; 0.1898; 0.1910
E_a^{21}	0.084 %	0.005 %	0.204 %
GCI_{fine}^{21}	0.46 %	0.007 %	0.12 %

Table 6: Temporal discretisation error ($r = 2$; subscripts: 1 = $\tau/200$, 2 = $\tau/100$, 3 = $\tau/50$).

Quantity	$\psi = Nu$	$\psi = T_{max}$ (K)	$\psi = \beta_s$
Δt_1; Δt_2; Δt_3 (s)		3.73×10^{-5}; 7.47×10^{-5}; 1.49×10^{-4}	
ψ_1; ψ_2; ψ_3	18.675; 18.689; 18.677	359.212; 359.201; 359.205	0.1898; 0.1898; 0.1888
E_a^{21}	0.071 %	0.003 %	0.006 %
GCI_{fine}^{21}	0.49 %	0.002 %	≈ 0 %

The uncertainty remains below 0.5 % for all quantities: the medium mesh and $\Delta t = \tau/100$ are retained.

7 Study design

Each parameter is varied independently, the other two being held at $\theta_0 = 10°$, $St = 0.3$ and $\varphi = 0.04$ (Table 7).

Table 7: Study design.

Parameter	Values
θ_0 (°)	5; 10; 15
St	0.1; 0.3; 0.5
φ	0; 0.02; 0.04

8 Results and discussion

The results of the seven simulations are summarised in Table 8.

Table 8: Results of the parametric study.

θ_0 (°)	St	φ	Nu/Nu_0	Δp (Pa)	PEC	PEC_{act}	T_{max} (K)	β_s
5	0.3	0.04	1.041	59.39	0.856	0.856	359.92	0.183
10	0.3	0.04	1.055	60.45	0.862	0.860	359.20	0.190
15	0.3	0.04	1.076	61.97	0.872	0.868	357.64	0.203
10	0.1	0.04	1.043	60.02	0.855	0.855	359.94	0.182
10	0.5	0.04	1.048	60.18	0.858	0.853	359.21	0.181
10	0.3	0	1.064	55.82	0.893	0.891	358.31	–
10	0.3	0.02	1.059	57.99	0.878	0.876	358.76	0.201

8.1 Heat transfer gain and hydraulic cost

The airfoils improve heat transfer only modestly, whereas the pressure drop is nearly doubled relative to the smooth channel (Table 8).

The pressure drop is set by the blockage of the two inclined airfoils: the oscillation amplitude and frequency change it only slightly. Heat transfer, in contrast, depends mainly on the amplitude, which widens the wake and enhances mixing near the wall.

Among the frequencies tested, $St = 0.3$ gives the best heat transfer. At $St = 0.5$, the transverse agitation measured downstream (v_{rms}) is lower than at $St = 0.3$.

Finally, NEPCM increase the viscosity of the mixture, and hence the pressure drop, without improving the Nusselt number.

8.2 Overall performance

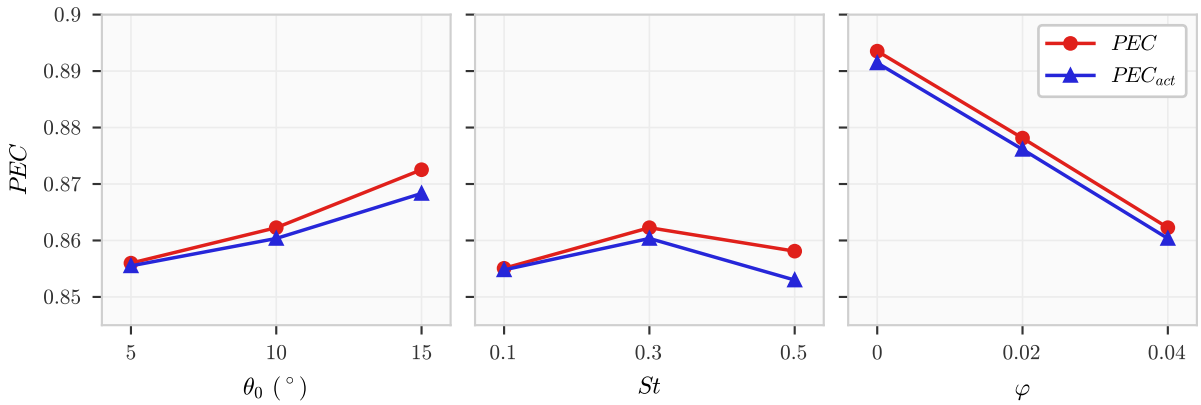


Figure 3: PEC and PEC_{act} as functions of each parameter.

The PEC remains below 0.9 in all cases (Fig. 3): the heat transfer gain does not offset the pressure drop. It increases with the amplitude, peaks at $St = 0.3$ and decreases with the NEPCM fraction. The actuation power remains negligible compared with the pumping power, hence $PEC_{act} \approx PEC$: the cost comes from the presence of the airfoils, not from their motion.

8.3 Maximum temperature

The reduction in maximum temperature remains limited, of the order of a few kelvins relative to the smooth channel (Fig. 4). The amplitude is the only parameter that reduces it noticeably. NEPCM have the opposite effect: the maximum temperature increases with φ.

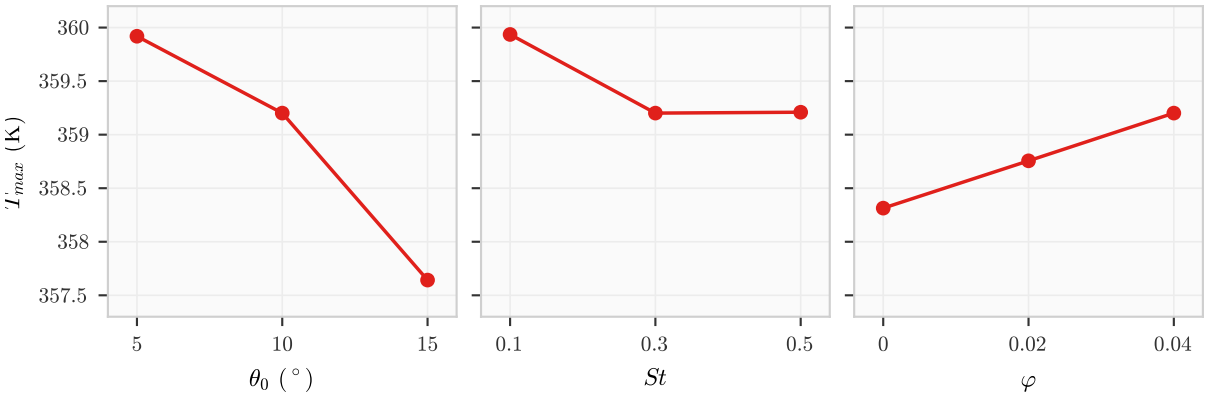


Figure 4: Maximum wall temperature as a function of each parameter.

8.4 Velocity, temperature and specific heat fields

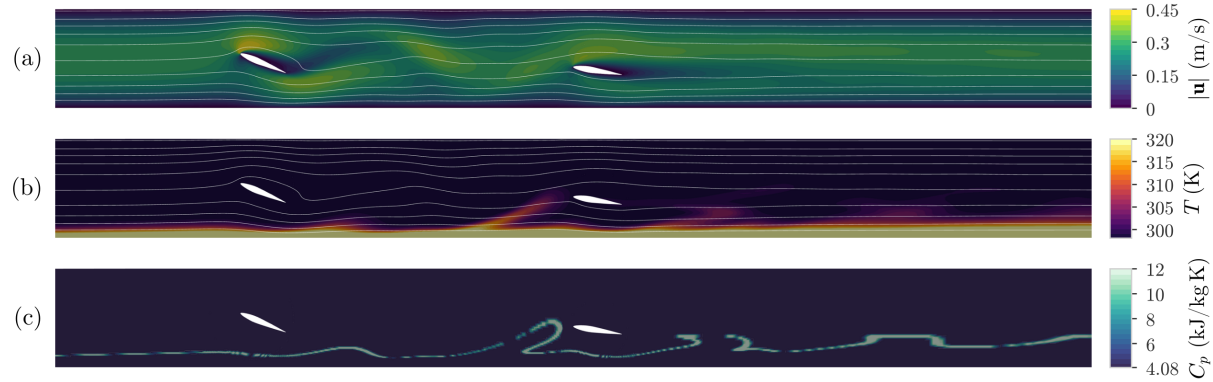


Figure 5: Instantaneous fields for $\theta_0 = 15°$, $St = 0.3$, $\varphi = 0.04$: (a) velocity and streamlines, (b) temperature, (c) specific heat of the mixture.

Airfoil 1 produces a wavy wake that airfoil 2 deflects towards the heated wall (Fig. 5). Heat remains confined to a thin layer along the wall: the wake pulls hot plumes away from it, but the core of the flow stays close to T_{in}. The C_p field shows that melting occurs only within a narrow band, which follows the undulations of the wake.

8.5 Melt fraction

The melt fraction at the outlet varies little between cases and increases slightly with the amplitude, as stronger mixing brings more capsules into the hot region. It remains close to 20 %: most of the latent heat carried by the capsules is not used.

The melting range lies only a few kelvins above T_{in}. Capsules near the wall therefore melt early and store no further heat, while those in the core, which remains cold, do not melt. Yet the capsules raise the viscosity throughout the channel: their hydraulic cost is paid everywhere, whereas their benefit is obtained only near the wall.

9 Conclusions

1. At equal velocity, the airfoils are not beneficial: the PEC remains below 0.9 in all cases. In this laminar micro-channel, they cost more in pumping power than they bring in heat transfer.

2. The pressure drop is set by the presence of the airfoils, not by the oscillation, whose actuation power is negligible.

3. The amplitude is the most effective parameter; among the frequencies tested, $St = 0.3$ performs best.

4. NEPCM degrade performance: about 20 % of the capsules melt, and their effect on viscosity dominates.

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